Concept Drift, Continuous ML, MLOps & CCAM: A Drift-aware Platform

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Agenda

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Thesis Scope

- Build a **drift-aware platform** for mobility models on Athens center.
- Generate **synthetic traffic data** from SUMO (Simulation of Urban MObility).
- **Three regression tasks**: ETA, Fuel Consumption, Number of Stops.
- Show end-to-end: **stable** $\rightarrow$ **drift** $\rightarrow$ **detect** $\rightarrow$ **retrain** $\rightarrow$ **mitigate**.
- Deliverables: dataset generation pipeline, tuned models for each task, platform (backend, frontend, drift service, model services).

Concept Drift & Why It Matters

- **What it is:** When the real world changes, the link between inputs and the target also changes over time (non-stationarity).
- **Why it happens:** Seasons and trends, one-off events, policy or product changes, and even sensor or software updates.
- **Why it matters:** If we don't watch for it, model accuracy quietly drops, dashboards stop matching reality, and decisions become costly or unsafe.
- **What to do:** Treat ML as a live system—**monitor** data and outcomes, **detect** changes early, and **adapt**.

Our ML Tasks

- **Estimated Time of Arrival (ETA)**: essential for accurate route planning and improving user satisfaction.
- **Fuel Consumption**: key to reducing operating costs and CO_2 emissions via eco-routing.
- **Number of Stops**: influences perceived trip quality and helps optimize overall traffic flow.

Software Design & Tools

- **Synthetic Data**: SUMO - Python.
- **Models**: XGBoost/LightGBM/CatBoost, scikit-learn, numpy, pandas - Python.
- **Backend**: FastAPI - Python.
- **Drift**: River (ADWIN, KSWIN, PageHinkley) - Python.
- **UI**: Dash/Plotly - Python.
- **Data**: Parquet/CSV files.
- **Orchestration**: Docker.

Athens Network Configurations & Traffic Patterns

- **Base network**: SUMO network built from OSM data over central Athens.
- **Rain network**: same map area, but with friction reduced in all lanes ($1.0 \rightarrow 0.4$) to simulate wet conditions.
- **Traffic generation periods**: per-hour volumes with morning and afternoon peaks, and lower mid-day activity.
- **Realistic variability**: added random noise to hourly volumes and used a distinct random seed for each 10h day simulation.

Dataset Structure

- **Three CSV files** (each ∼55–60 k trips over a 10 hour simulation):
  - `train-fcd.csv` (base network)
  - `test-fcd.csv` (base network)
  - `rain-fcd.csv` (rain-modified network)
- **Columns (per row)**: `timestep, vehicle_id, x, y, speed, lane, odometer, fuel, waiting_time`

Dataset Generation Pipeline

- **osmGet**: extract OSM data within a bounding box over central Athens
- **osmBuild**: convert OSM into a SUMO-compatible network file
- **randomTrips**: generate vehicle trips using per-hour traffic volumes and a different random seed
- **sumo**: run 10-hour simulations, export Floating Car Data (FCD) and emission logs

Data Preprocessing & Feature Engineering

- **Trip extraction**: group raw FCD by vehicle into one row per trip (origin-destination)
- **Different features** added per model:
  - *Temporal*: `hour_bin`
  - *Spatial*: source/destination clusters (`x`,`y`)
  - *Route*: number of edges traversed, distance measures (Euclidean, Manhattan, Haversine)
- **Result**: tabular dataset with inputs + ground-truth targets (ETA, fuel, stops)

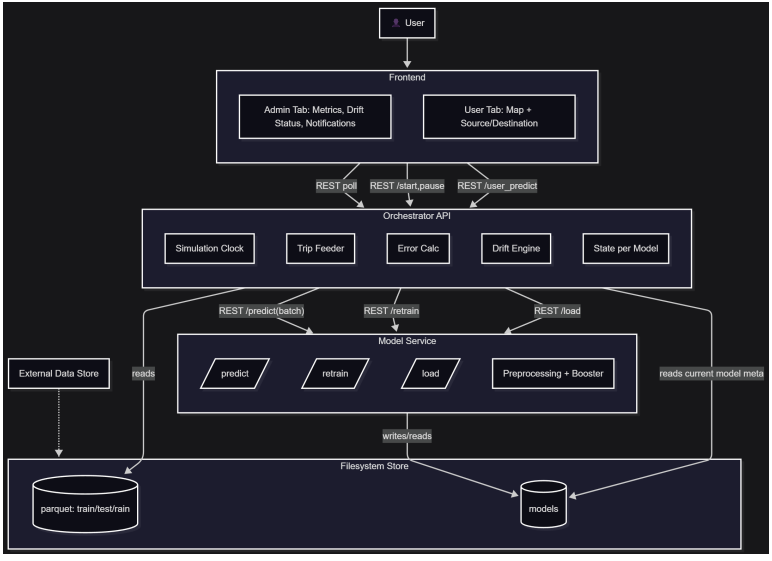
Model Training & Evaluation

- **Evaluation metrics**:
  - *Primary*: MAE
  - *Secondary*: MAPE, R^2, RMSE, MSE
- **Hyperparameter tuning**: Optuna / grid search over `n_estimators`, `max_depth`, `learning_rate`, `subsample`, ...
- **Outcome**: XGBoost delivered the best trade-off of accuracy and training speed for all three tasks

High-level Architecture

- **Frontend (Dash/Plotly)**: Admin tab (live MAE charts, notifications, run controls); User tab (Athens map, O/D query).
- **Orchestrator (FastAPI)**: simulation clock, batch prediction, error calc, drift ensemble, per-model state, UI feed, controls.
- **Model Service (FastAPI)**: per-task pipelines (preprocessing + booster), `/predict`, `/retrain`, `/load`.
- **Filesystem Store**: Parquet datasets (`train/test/rain`); filesystem model registry (`models/{task}/vN/`).
- **Docker Compose**: orchestration with multiple containers.

Components Diagram



How It Works (at a glance)

1. **Timelapse clock**: compress 20 h of trips (test then rain) into a few minutes.
2. **Per tick**: Orchestrator pulls trips whose `sim_time` falls in the current window.
3. **Batch predict**: send features to Model Service; receive predictions for ETA/Fuel/Stops.
4. **Immediate ground truth**: join with labels; compute per-trip absolute errors.
5. **Live metrics**: update rolling MAE per task and push to the UI feed.
6. **Drift monitoring**: feed a smoothed error stream into ADWIN, Page–Hinkley, KSWIN, and an SPC rule; raise drift on majority vote.
7. **Mitigation**: after drift is confirmed, collect N more "rain" trips, trigger retraining, then swap to the new model version.
8. **Repeat per model**: ETA, Fuel, Stops progress through the same state machine independently.

Admin Tab: Sped-up Simulation & Drift Story

- **Three live charts** (ETA/Fuel/Stops): rolling MAE; drift state (Stable/Confirmed/Collecting/Retraining/Swapped).
- **Notifications**: day transition at 10 h (*rain starts*), drift detected (*votes* ≥ 3), retrain started/completed, model swapped to vN.
- **What you see**: errors rise after rain; drift fires; we keep predicting with old model while "collecting" rain data; retrain runs; swap reduces error (not all the way to baseline).
- **Controls**: Start/Pause/Resume/Restart.

Admin Tab: Sped-up Simulation & Drift Story

User Tab: Predictions at the Current State

- **Map of central Athens**: user picks **source** and **destination**.
- **Uses the same simulation time**: features derive from current `sim_time` (e.g., `hour_bin`, distance, spatial bins).
- **Calls active models**: returns ETA, Fuel, Stops from whichever version is currently loaded (pre- or post-swap).
- **Consistent context**: shows badges like "Rain scenario" and current drift state so predictions match what the Admin tab shows.
- **Demo behavior**: *pauses* the timelapse while in the User tab, then *resumes* when you go back to Admin.

User Tab: Predictions at the Current State

Implementation Challenges

- **Road closures drift scenario:** Closing lanes/edges to mimic events/roadworks in SUMO led to unstable routing and distorted traffic patterns. We therefore used the rain/friction shift as a controlled, repeatable drift.
- **Parking availability task:** The current SUMO setup lacks realistic parking supply/occupancy and consistent stop events, so the task could not be integrated or evaluated alongside the trip-level models.
- **Time-series traffic speed model:** A lagged-input traffic speed model adapted quickly to the rain shift and showed little error increase, making drift hard to showcase; we replaced it with the "number of stops" task which reacts clearly.

Questions

Thank you!